

SHRUTI JAIN

+1-412-589-5683 | spjain@andrew.cmu.edu | [in shruti-jain-1405](https://www.linkedin.com/in/shruti-jain-1405) | [shrutijain1405.github.io](https://github.com/shrutijain1405)

EDUCATION

Carnegie Mellon University | Master of Science in Computer Vision Aug 2025 - Expected Dec 2026
Coursework: Learning for 3D, Geometry based Vision, Advanced Computer Vision, Introduction to Machine Learning
CGPA: 4.17/4

Manipal Institute of Technology | Bachelor of Technology in Information Technology 2019 - 2023
CGPA: 3.48/4

SKILLS

Programming Languages/Tools: Python, C++, Java, Docker, Kubernetes, SQL, Temporal
ML and CV Frameworks/Libraries: PyTorch, OpenCV, ROS, CUDA, TensorFlow, AWS
AI Topics: Multimodal Learning, Vision-Language Models, Diffusion Models, Post-Training (LoRA, GRPO), Object Tracking and Re-ID, 3D Gaussian Splatting,

RELEVANT EXPERIENCE

AIRLab - Carnegie Mellon University Jan 2026 - Present
Research Assistant — Prof. Sebastian Scherer and Prof. John Galeotti

- Developing a **multiview thermal and near-infrared fusion framework** for nighttime UAV casualty assessment, combining semi-automated Re-ID with VLM-based posture and hemorrhage analysis.
- Developing a **scene graph** based approach to **cross-view, cross-modal casualty re-identification**, combining tracklet appearance, posture, and environmental context across aerial thermal and ground near-IR video.
- Built a two-stage system combining **diffusion-based RGB-to-IR generation** and VLM fine-tuning with **LoRA/GRPO optimization** to improve triage reasoning under limited paired RGB-IR data.

FlowState AI — Computer Vision Research Intern May 2026 - Aug 2026

- Architected and shipped an **end-to-end AI video-editing system** from transforming long-form footage into prompt-driven, story-based vertical content with intelligent 16:9-to-9:16 reframing.
- Built a **self-correcting video pipeline** combining **VLM** subject identification and grounding, **SAM3** tracking, and trajectory verification, with scene and shot-level orchestration in **Temporal**.
- Isolated and evaluated individual pipeline components, achieving **15.4% increase in precision** for Soccer ball tracking by introducing automated jerk detection and correction in soccerball trajectories.

Carnegie Mellon University, USA Aug 2025 – Dec 2025
Research Assistant — Prof Deva Ramanan

- Researched and developed DetPO, a gradient-free black-box prompt optimization framework achieving up to **9.7% improvement** over prior approaches on Roboflow20-VL and LVIS.

Palo Alto Networks — Software Developer Oct 2023 - Apr 2025

- Worked on features and APIs of the microservices (**Java Spring Boot**) for ingesting events from tenants' cloud accounts (**AWS, Azure, GCP**) for near real-time threat detection using **anomaly detection**.

Robotics Research Centre (RRC) - IIIT Hyderabad Jun 2023 – Sept 2023
Research Assistant — Prof. K Madhava Krishna

- Achieved **14.4%** average decrease in the mean absolute scores across 3 datasets as compared to the SOTA for Image Segmentation in **self-driving cars in adverse weather conditions**.
- Implemented a **differentiable visual and latent prompting mechanism** to fine-tune on **Segment Anything Model** via **adaptors** through learnable image pre-processing blocks.

SELECTED PUBLICATIONS

- DetPO: In-Context Learning with Multi-Modal LLMs for Few-Shot Object Detection, ECCV '26
- DiffPrompter: Differentiable Implicit Visual Prompts for Semantic-Segmentation in Adverse Conditions, IROS '24
- A Studious Approach To Semi-Supervised Learning, ICBINB Workshop, NeurIPS '21

RELEVANT PROJECTS

- Gated Multimodal Web Navigation Agents** - A gated multimodal web navigation framework combining MindAct and Qwen2-VL on the Multimodal Mind2Web benchmark, improving robustness to adversarial popup attacks.
- Editing 3D Gaussian Splats** - Built a **text-guided 3D Gaussian Splat editing** pipeline integrating **DDIM inversion** and **InstructPix2Pix** with **cross-view attention** to preserve multi-view consistency, supporting global scene edits and localized object edits through language-guided segmentation masks.
- Video Translation with LipSync** - Built an **end-to-end English-to-German video dubbing pipeline** integrating translation, voice-cloned speech synthesis, audio timing alignment, and LatentSync **lip synchronization**.
- Real-time Failed 3D Print Detector** - Detects failed 3D prints in real time using OpenCV and **anomaly detection**.